

Mathematical Foundations of Political Methodology

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- Sequence
- Limits of Sequence
- Cauchy Sequence
- Limits of Functions
- Continuity

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Sequence

Sequence can be irregular, an example of such sequence is:

$$a_1 = 1$$

$$a_2 = 8$$

$$a_3 = 6$$

$$a_4 = \dots$$

Sequence can be regular, an example of such sequence is:

$$x_1 = 2$$

$$x_2 = 4$$

$$x_3 = 6$$

$$x_4 = \dots$$

Sequence

Definition

A *sequence* of real numbers is a function, u , mapping $\mathbb{N}$ to $\mathbb{R}$

- We will be dealing with infinite sequences
- Written $u(1), u(2), u(3), \dots$, or $(u_1, u_2, u_3, \dots)$
- The sequence is written $\{u_n\}$
- Example of a sequence

$$u_n = \frac{3n^2}{n^3 + 9}$$

which gives the sequence

$$\frac{3}{10}, \frac{12}{17}, \frac{48}{73}, \dots$$

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Arithmetic Sequences

Definition

An arithmetic sequence is a sequence $\{u_n\}$ in which

$$u_{n+1} - u_n$$

is constant for all n

- Equivalent to saying that for some numbers a, d , the sequence looks like

$$a, a + d, a + 2d, a + 3d, \dots$$

or

$$u_n = a + (n - 1)d$$

- Is this sequence arithmetic? (If so, what are a, d ?)

$$u_n = 14 - 5n$$

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Geometric sequences: Each term is obtained by multiplying previous term by same number.

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Limits of Sequence: Motivation

Now, we want to ask a question, as n goes further and further, that is, n goes to infinity, what is the behavior of the sequence?

- Sequence
- Limits of Sequence
- Cauchy Sequence
- Limits of Functions
- Continuity

Limits of sequences

Definition

A sequence $\{u_n\}$ tends to the limit z if for any positive real number ϵ there is a positive integer, N , such that:

$$|u_n - z| < \epsilon \text{ for all } n > N.$$

Also written as

$$\lim u_n = z$$

$$\lim_{n \rightarrow \infty} u_n = z$$

$$u_n \rightarrow z \text{ as } n \rightarrow \infty$$

Limits of Sequences

■ Some Remarks

- If a sequence converges, it converges to one number. We call that A
- $\epsilon > 0$ is some arbitrary real-valued number.
- The N will depend upon ϵ
- Implies the sequence never gets further than ϵ away from A

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Limits of Sequences

Theorem

$\{\frac{1}{n}\}$ converges to 0

Proof.

We need to show that for ϵ there is some N_ϵ such that, for all $n \geq N_\epsilon$ $|\frac{1}{n} - 0| < \epsilon$. Without loss of generality (WLOG) select an ϵ . Then,

$$\begin{aligned} \left| \frac{1}{N_\epsilon} - 0 \right| &< \epsilon \\ \frac{1}{N_\epsilon} &< \epsilon \\ \frac{1}{\epsilon} &< N_\epsilon \end{aligned}$$

For each epsilon, then, any $N_\epsilon > \frac{1}{\epsilon}$ will suffice. □

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Cauchy Sequence: Motivation

A fun fact, how do we pronounce Cauchy, KohShee.

In previous example, we show $\frac{1}{n}$ converges to 0 because we know its limit. However, a natural follow-up question to ask is that can we show that a sequence has limit, but without knowing the limit ex ante.

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- **Cauchy Sequence**
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Divergent Sequence

Definition

A sequence that tends to a limit is called convergent; a sequence that doesn't is called divergent.

- A sequence that tends to infinity (or negative infinity) is divergent, but a sequence doesn't have to tend to infinity to be divergent. (What's an example of a divergent sequence that doesn't tend to infinity?)

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Definition

$\{x_n\}$ is Cauchy if $\forall \epsilon > 0, \exists N \in \mathbb{N} : |x_m - x_n| < \epsilon$ when $m, n > N$

Theorem

In $\mathbb{R}$, $\{x_n\}$ is convergent if and only if it's Cauchy.

Apply Cauchy Criterion

We can show the sequence defined by $x_n = n$ is not convergent:

$|x_n - x_m| = |n - m| \geq 1$ for all n, m . Thus, for $\epsilon = 1$ there is no N such that $|x_n - x_m| < \epsilon$ whenever $n, m > N$. Therefore $\{x_n\}$ is not Cauchy, and does not tend to a limit.

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Working with Limits

- Fact 1: For a real number x , $\lim_{n \rightarrow \infty} x^n = 0$ if and only if $-1 < x < 1$.
- Fact 2: For a real number b , $\lim_{n \rightarrow \infty} n^b = 0$ if and only if $b < 0$.

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Working with Limits: Evaluate Dominating Term

What is the limit of

$$u_n = \frac{(-1)^n(n+1)^2}{n^4}$$

Hint: We can start by expanding the numerator.

What would happen if we replaced $(n+1)^2$ with $(n+1)^4$?
Don't do whole calculation!

Another practice, though easier one, as $n \rightarrow \infty$, what's the limit of

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Series

- The series defined by a sequence $\{u_n\}$ is the sequence $\{s_n\}$ with

$$s_1 = u_1, s_2 = u_1 + u_2, s_3 = u_1 + u_2 + u_3, \dots$$

and more generally

$$s_n = \sum_{r=1}^n u_r$$

- Thus, series is a special type of sequence, it sums the terms in the sequence.

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- Thus, series is a special type of sequence, it sums the terms in the sequence.

Formula for Summing Algebraic Sequence

We are interested in the sum of first n terms in arithmetic sequence. Here is the derivation:

First, the natural numbers

$$s_n = 1 + 2 + \dots + (n - 1) + n$$

$$s_n = n + (n - 1) + \dots + 2 + 1$$

Adding these together we get

$$2s_n = (1 + n) + (1 + n) + \dots + (1 + n) = n(1 + n)$$

The first n terms of the natural numbers sum to

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Formula for Summing Algebraic Sequence

Recall that a general arithmetic sequence is $u_n = a + (n - 1)d$

We can use the same trick to find the sum of the first n terms of the sequence:

$$u_n = a + (a + d) + (a + 2d) + \dots + (a + (n - 2)d) + (a + (n - 1)d)$$

$$u_n = (a + (n - 1)d) + (a + (n - 2)d) + \dots + (a + 2d) + (a + d) + a$$

$$2u_n = n(2a + (n - 1)d)$$

$$u_n = an + n \left(\frac{(n - 1)d}{2} \right)$$

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Summing Geometric Sequences

Summing geometric sequences: $s_n = a + ax + ax^2 + \dots + ax^{n-1}$

Multiplying by x we get

$$xs_n = ax + ax^2 + \dots + ax^n$$

Let

$$t = ax + ax^2 + \dots + ax^{n-1}$$

Then $s_n = a + t$ and $xs_n = t + ax^n$

$$s_n - xs_n = a - ax^n$$

so

$$s_n(1 - x) = a - ax^n,$$

and finally

$$s_n = a \left(\frac{1 - x^n}{1 - x} \right)$$

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An Example for Summing Geometric Sequence

Now suppose you save 1000 dollars in Citibank at year 0 (that is, the beginning year). After year 0, you will save 1000 dollars at Citibank every year. And the annual (compound) interest rate is 5 percent.

Then, a natural question will be: at year n , how much money do you make.

For the money you saved at year 0, you will get: $1000 * (1 + 0.05)^n$;

For the money you saved at year 1, you will get: $1000 * (1 + 0.05)^{n-1}$;

For the money you saved at year 2, you will get: $1000 * (1 + 0.05)^{n-2}$;

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Sum them together will give you the money you make:

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An Example for Summing Geometric Sequence

Now suppose you save 1000 dollars in Citibank at year 0 (that is, the beginning year). After year 0, you will save 1000 dollars at Citibank every year. And the annual (compound) interest rate is 5 percent.

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Convergence of Geometric Series

We will often care about the limit of a series as $n \rightarrow \infty$

For a geometric series,

$$s_n = a \left(\frac{1 - x^n}{1 - x} \right)$$

We know that $x^n \rightarrow 0$ as $n \rightarrow \infty$ if and only if $|x| < 1$

If $|x| < 1$ then $\lim_{n \rightarrow \infty} s_n = a \left(\frac{1-0}{1-x} \right) = \frac{a}{1-x}$

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Divergence of Geometric Series

Recall that $s_n = \sum_{k=1}^n k_n$.

Fact (*term test* for any series): If $\lim_{n \rightarrow \infty} k_n \neq 0$ then $\sum_{n=1}^{\infty} k_n$ diverges

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Example in Multiperiod Utility Function

Suppose that I plan to consume x units of cake in each of an infinite number of time periods, $t = 1, 2, 3, \dots$

I discount the utility I'll receive from future consumption by a term $0 < \delta < 1$, so that I value today's consumption fully as $u(x)$, tomorrow's as $\delta u(x)$, the day after's as $\delta^2 u(x)$...

Now, let's see a concrete example, let's set $\delta = 0.9$. Then, tomorrow's utility is: $0.9 * u(x)$, the day after's as $0.9^2 * u(x)$...

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My expected utility over time is to sum my utilities together:

$$\delta^0 u(x) + \delta^1 u(x) + \delta^2 u(x) + \delta^3 u(x) + \dots$$

We can write this summation in a compact way:

$$\sum_{t=0}^{\infty} \delta^t u(x) = \frac{u(x)}{1 - \delta}$$

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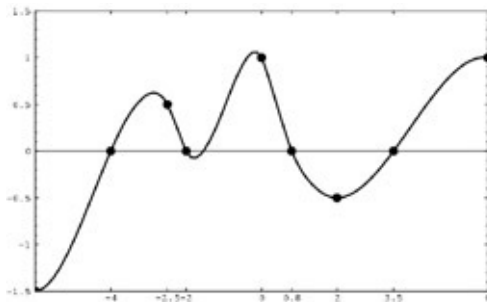
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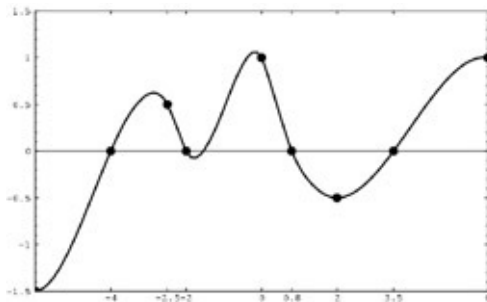
Limits of Functions: Motivation



We care about the limits of function because we care about how does function behave at a point. Essentially, this is the most important aspect in calculus (more broadly to say, real analysis): the study of limit.

And the more calculus and real analysis you learn, you will realize the importance of limits.

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And the more calculus and real analysis you learn, you will realize the importance of limits.

Limits of Functions

Definition

Suppose $f : R \rightarrow R$. We say that f has a limit L at x_0 if, for each $\epsilon > 0$, there is a $\delta > 0$ such that $|x - x_0| < \delta$ implies that $|f(x) - L| < \epsilon$.

- Limits are about the behavior of functions at points. Here x_0 .
- As with sequences, we let ϵ define an error rate
- δ defines an area around x_0 where $f(x)$ is going to be within our error rate

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Limits of Functions: Example

Theorem

The function $f(x) = \frac{x^2-1}{x-1}$ has a limit of 2 at $x_0 = 1$.

Proof.

For all $x \neq 1$,

$$\begin{aligned}\frac{x^2 - 1}{x - 1} &= \frac{(x + 1)(x - 1)}{x - 1} \\ &= x + 1\end{aligned}$$

Choose $\epsilon > 0$ and set $x_0 = 1$. Then, we're looking for δ_ϵ such that

$$|x - 1| < \delta_\epsilon \quad \text{implies} \quad |(x + 1) - 2| < \epsilon$$

Again, if $\delta_\epsilon = \epsilon$, then this is satisfied. □

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Remarks on Limits of Functions

$\lim_{x \rightarrow x_0} f(x)$ might or might not equal $f(x_0)$

- It could be *different than* $f(x_0)$

$$f(x) = \begin{cases} +1 & \text{if } x \neq 4 \\ -1 & \text{if } x = 4 \end{cases}$$

- It could be *undefined*

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Algebra of Limits

Theorem

Suppose $f : \mathbb{R} \rightarrow \mathbb{R}$ and $g : \mathbb{R} \rightarrow \mathbb{R}$ with limits A and B at x_0 . Then,

$$i.) \lim_{x \rightarrow x_0} (f(x) + g(x)) = \lim_{x \rightarrow x_0} f(x) + \lim_{x \rightarrow x_0} g(x) = A + B$$

$$ii.) \lim_{x \rightarrow x_0} f(x)g(x) = \lim_{x \rightarrow x_0} f(x) \lim_{x \rightarrow x_0} g(x) = AB$$

Suppose $g(x) \neq 0$ for all $x \in \mathbb{R}$ and $B \neq 0$ then $\frac{f(x)}{g(x)}$ has a limit at x_0 and

$$\lim_{x \rightarrow x_0} \frac{f(x)}{g(x)} = \frac{\lim_{x \rightarrow x_0} f(x)}{\lim_{x \rightarrow x_0} g(x)} = \frac{A}{B}$$

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Continuity

Definition

Suppose $f : \mathbb{R} \rightarrow \mathbb{R}$ and consider $x_0 \in \mathbb{R}$. We will say f is continuous at x_0 if for each $\epsilon > 0$ there is a $\delta > 0$ such that if,

$$\begin{aligned} |x - x_0| &< \delta \text{ for all } x \in \mathbb{R} \text{ then} \\ |f(x) - f(x_0)| &< \epsilon \end{aligned}$$

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